



How higher education affects corporate ESG performance: Empirical evidence from Chinese listed companies

Kaidi Tu^{a,b}, Yuanyuan Guo^{b,c,*}

^a School of Administration, Jilin University, 130012, China

^b Henan Mechanical and electrical vocational College, 451191, China

^c School of Education, Central China Normal University, 430079, China

ARTICLE INFO

Keywords:

High education

Corporate ESG performance

ABSTRACT

This paper empirically examines the impact of higher education on corporate ESG (Environmental, Social, and Governance) performance among non-financial and non-real estate listed companies in China's capital market. The findings suggest that a higher level of education among corporate executives is conducive to improving corporate ESG performance levels. Mechanism tests indicate that the impact of higher education on corporate ESG performance is indirectly realized through the enhancement of internal control quality within companies. Heterogeneity analysis shows that the positive effect of higher education on improving corporate ESG performance is more pronounced in non-state-owned enterprises.

1. Introduction

Corporate ESG performance assessment is a comprehensive process that involves a thorough examination of a company's performance in environmental, social, and governance aspects. Environmental performance looks at a company's environmental protection measures, energy efficiency, waste management, and adaptation and mitigation measures against climate change. Social performance focuses on the relationship between the company and its employees, community, customers, and suppliers, including aspects such as employee welfare, human rights protection, community investment, and consumer rights protection (Zhang et al., 2023). Governance performance emphasizes the company's management structure, decision-making transparency, board independence, and anti-corruption measures to ensure fairness and transparency in company management. By comprehensively assessing a company's ESG performance, investors and stakeholders can gain a more complete understanding of the company's risks and opportunities, thereby better guiding investment decisions and promoting corporate development towards sustainability and social responsibility.

Higher education, defined as education provided after the secondary education stage, offering more in-depth, specialized knowledge and skills training, plays a crucial role in promoting corporate ESG performance. It usually involves universities, colleges, and professional institutions offering degree programs at various levels, such as bachelor's, master's, and doctoral degrees. The aim of higher education is to cultivate talent with professional knowledge, critical thinking, and innovative capabilities to meet the complex needs of modern society and the economy. The content of higher education spans various disciplines, including natural sciences, social sciences, humanities, engineering and technology, as well as interdisciplinary and comprehensive professional training. Higher

* Corresponding author.

E-mail address: 412859686@qq.com (Y. Guo).

education is considered a critical pillar of national development in countries around the world, playing an essential role in advancing societal progress, promoting economic growth, and nurturing talent.

Higher education plays a significant role in enhancing corporate ESG performance in several ways. First, it cultivates talents with a broad knowledge base and critical thinking skills, who are more inclined to focus on ESG issues and integrate them into corporate strategic planning and business decisions. Secondly, higher education institutions themselves are important venues for ESG research and innovation, driving development in relevant fields and providing new solutions and business models for companies. Third, higher education institutions actively promote the cultivation of social responsibility awareness, collaborating with companies on various projects and activities to explore ESG issues together, thus promoting improvement and development in environmental, social, and governance aspects. Lastly, higher education institutions provide consulting services and expertise on ESG matters, helping companies to formulate and implement strategies and measures that meet ESG standards. Overall, through talent cultivation, research and innovation, social responsibility, and consulting services, higher education positively influences and promotes the improvement of corporate ESG performance.

While existing research has predominantly explored the impacts of the financial development scale on aspects like income disparity, banking risks, and corporate investment within a company's region, the economic consequences of the financial geographic structure have been less examined. Furthermore, the integration of cultural factors into this discourse remains scant, leaving a gap in a comprehensive understanding of how financial geographic structure influences corporate economics. This manuscript aims to delve into the effects of higher education on corporate ESG performance among non-financial and non-real estate listed companies in China's capital market. By focusing on the educational background of corporate executives, this study explores the link between higher education and ESG performance, emphasizing the role of internal control quality as a mediating factor and examining the variability of these effects across different ownership types, particularly in non-state-owned enterprises.

The findings of this paper indicate that executives with higher education backgrounds play a significant role in enhancing corporate ESG performance, and mechanism tests show that higher educational backgrounds of executives can further promote ESG performance improvement through enhancing the quality of internal controls. Heterogeneity tests reveal that the positive effect of higher education on corporate ESG performance is more pronounced in non-state-owned enterprises.

The marginal contributions of this paper are threefold: firstly, it demonstrates the significant positive role of executives with higher education backgrounds in enhancing corporate ESG performance, highlighting the importance of higher education in shaping the quality of corporate leadership teams and improving ESG performance. This finding provides valuable insights for corporate recruitment and selection of executives, as well as supporting the value of higher education in talent development. Secondly, the study further explores the mechanism by which executives' educational backgrounds affect corporate ESG performance, revealing that higher educational backgrounds are conducive to improving the quality of internal controls, thereby further enhancing corporate ESG performance. This deepens the understanding of the mechanisms by which higher education affects corporate ESG performance, offering more specific guidance for corporate management practices. Lastly, the study's heterogeneity test results suggest that in non-state-owned enterprises, the positive effect of higher education on corporate ESG performance is more pronounced. This finding indicates that the impact of higher education on corporate ESG performance varies among different types of enterprises, helping to understand the moderating role of corporate governance structure and ownership nature on the mechanisms of higher education's impact, and providing more precise guidance for corporate practice.

2. Literature review

ESG, originating from responsible investment, stands for Environmental, Social, and Governance. It extends and enriches the concepts of the green economy, corporate social responsibility (CSR), and responsible investment (Li et al., 2022; Liu et al., 2023). The Principles for Responsible Investment (PRI) define responsible investment as "strategies and practices that incorporate Environmental, Social, and Governance (ESG) factors into investment decision-making and active ownership." ESG, as a sustainable and coordinated development value, balances economic, environmental, social, and governance benefits, aiming for long-term value growth. Unlike CSR, which indirectly includes governance issues related to environmental and social considerations, ESG explicitly encompasses these governance issues, focusing on the non-financial aspects and pursuing the company's long-term value by considering the environment, society, and corporate governance, reflecting sustainable development and green values (Wang et al., 2024; Gillan et al., 2021). The performance in environmental, social, and governance (corporate ESG performance) impacts capital costs (Tang, 2022; Nazir et al., 2022), investment efficiency (Wang et al., 2022), financial constraints (Wu et al., 2022), and corporate risk (Broadstock et al., 2021; Gao et al., 2022), among others. However, few studies have focused on the impact of higher education levels within corporations on corporate ESG performance levels.

3. Data and variables

3.1. Data sources

This paper selects all non-financial and non-real estate listed companies in China's Shanghai and Shenzhen A-share capital markets from 2013 to 2022 as the initial research sample. The initial sample was screened according to the following procedure: (1) exclude ST and PT samples; (2) exclude companies with liabilities exceeding assets; (3) to ensure the validity of the research sample, exclude samples with corporate income tax rates greater than 1 or less than 0; (4) exclude samples missing key variables; (5) apply Winsorizing to all continuous variables at the 1 % to 99 % levels. Data on executives' higher education backgrounds mainly come from the Wind

Table 1
Descriptive statistics.

VarName	Obs	Mean	SD	Min	Median	Max
Degree	17,032	3.6223	0.9552	1	4	5
ESG	17,032	5.3256	1.0263	3	6	9
Size	17,032	22.5803	1.0513	18.2815	23.8733	26.6372
Lev	17,032	0.4182	0.1835	0.0341	0.5709	0.8204
Roa	17,032	0.2192	0.7171	−0.6323	0.1367	14.3
Growth	17,032	0.1424	0.2415	−0.4284	0.5764	1.1375
PerIncome	17,032	0.6439	0.7208	−1.2128	0.4277	5.6061
OwnBalance	17,032	0.0758	0.09175	−0.6617	0.0649	0.9686
Tan_Rate	17,032	0.2949	0.3179	0.1135	0.2747	0.6094
Insinvestorprop	17,032	0.1924	0.1191	0.0155	0.1811	0.7124

database, while other financial data are mainly sourced from the CSMAR database.

3.2. Variables

3.2.1. Higher education

Higher education is widely regarded as one of the best ways to shape individuals. Individuals with higher education not only have advantages in professional competence but also show significant strengths in various aspects. Therefore, corporate executives with high-level educational training can make rapid and accurate decisions in complex and changing market environments and demonstrate relatively strong resilience. This paper's explanatory variable is the education level of executives. After summarizing a large amount of relevant literature and considering the availability of data, this paper divides the education level of senior executives into five levels (Degree) from low to high: junior college and below, associate degree, bachelor's degree, master's degree, and doctorate and above, assigned values from 1 to 5 respectively.

3.2.2. ESG rating data

ESG ratings cover environmental, social, and corporate governance aspects, comprehensively assessing a company's performance in social impact dimensions. This includes employee welfare, community support, diversity, and inclusiveness, among others, helping to more accurately measure a company's social impact. Based on this, this paper uses the China Securities ESG rating index to measure the comprehensive performance of companies in the social impact dimension. Specifically, since the China Securities ESG rating includes nine levels (from AAA to C), following existing research (Dend et al., 2023), this paper assigns a value of 1 to the lowest level (C), with each higher level receiving an additional point, up to 9 points for the AAA level, to quantify the ESG performance level of the

Table 2
Higher education and corporate human capital investment.

	ESG	
	(1)	(2)
Degree	0.0016* (1.9426)	0.0014* (1.8143)
Size		0.0161** (2.2466)
Lev		−0.0158*** (−3.8327)
Roa		0.0151*** (2.6780)
Growth		−0.0081 (−1.2388)
PerIncome		−0.0174** (−2.3596)
OwnBalance		−0.0156 (−1.0120)
Tan_Rate		0.0080 (0.3160)
Insinvestorprop		−0.0318* (−1.7860)
Constant	−1.0735*** (−10.1371)	−1.0741*** (−10.2129)
Year	Yes	Yes
Industry	Yes	Yes
N	17,032	17,032
R-Square	0.0825	0.1933

Note: The numbers in brackets are t-values; ***, **, and * indicate significance levels at 1 %, 5 %, and 10 %, respectively.

research samples.

3.2.3. Control variables

The control variables selected for this paper include: company size (Size), represented by the natural logarithm of total business revenue; debt-to-asset ratio (Lev), represented by the proportion of total liabilities to total assets; growth ability (Growth), measured by the growth rate of net profit, calculated as the current year's net profit minus the previous year's net profit divided by the previous year's net profit; institutional investor shareholding proportion (Insinvestorprop), measured by the ratio of institutional investor shareholding to total shares; earnings per share (PerIncome), equal to net profit divided by the number of common shares; return on assets (Roa), measured by the proportion of net profit to total assets; equity balance (OwnBalance), measured by the ratio of the shareholdings of the second to the tenth largest shareholders to the shareholdings of the largest shareholder; and the intensity of fixed assets (Tan_rate), represented by the ratio of the net value of fixed assets to total assets.

4. Empirical results

4.1. Descriptive statistics

Table 1 reports the descriptive statistics of the main variables in this paper. The proxy variable for the level of higher education among executives in the sample (Degree) has a minimum value of 1 and a maximum value of 5, indicating a wide range of educational levels among executives in the study sample. The proxy variable for the ESG performance of the sample companies (ESG) has a minimum value of 3 and a maximum value of 9, with an average value of 5.3256 and a median of 6. This shows a significant variance in the ESG performance levels among the companies in the study sample, with an overall low level of ESG performance. The descriptive statistical results of the other variables are similar to existing studies and are not repeated here.

4.2. Baseline results

To investigate how higher education affects corporate ESG performance, this paper constructs the following econometric model:

$$ESG_{it} = \alpha + \beta_1 Degree_{it} + \gamma Controls_{it} + \delta_I + \lambda_Y + \varepsilon_{it} \quad (1)$$

In the model, δ_I and λ_Y represent industry fixed effects and year fixed effects, respectively, ε_{it} is the random disturbance term, and $Controls_{it}$ represents the control variables selected in the model, with specific interpretations as previously described. β_1 is the parameter of primary interest in this study. If β_1 is significantly positive, it indicates that a higher level of higher education among corporate executives is associated with a higher level of corporate ESG performance.

The empirical results of Model (1) are presented in Table 2. Column (1) shows the results controlling only for year fixed effects and industry fixed effects, while Column (2) shows the regression results after including a series of control variables. Overall, the regression coefficient between the proxy variable for the level of higher education received by corporate executives and the proxy variable for corporate ESG performance is significant at the 10 % level. This suggests that a higher level of education among corporate executives is conducive to enhancing corporate ESG performance levels, indirectly reflecting that higher education is beneficial in promoting the improvement of corporate ESG performance.

4.3. Robustness test

4.3.1. Two-Way fixed effects model

The ESG performance level of a company is influenced not only by industry common factors but also by the company's own unobservable individual characteristics. Therefore, this paper further adopts a two-way fixed effects model, which controls for both firm-specific and year-specific fixed effects on top of Model (1), to re-examine the relationship between higher education and corporate ESG performance. The test results, as shown in Table 3, reveal that the regression coefficient between the proxy variable for the level of

Table 3
Two-way fixed effects model.

	ESG	
	(1)	(2)
Degree	0.0271** (2.0923)	0.0254** (2.3321)
Constant	−2.0601*** (−6.6068)	−1.7098*** (−6.4314)
Controlvariable	No	Yes
Year	Yes	Yes
Firm	Yes	Yes
N	17,032	17,032
R-Square	0.1201	0.1302

Note: The numbers in brackets are t-values; ***, **, and * indicate significance levels at 1 %, 5 %, and 10 %, respectively.

Table 4

Excluding the impact of exogenous shocks.

	ESG	
	(1)	(2)
Degree	0.0562** (2.0442)	0.0343*** (2.6983)
Constant	1.0147*** (5.1847)	1.0151*** (5.3032)
Controlvariable	No	Yes
Year	Yes	Yes
Industry	Yes	Yes
N	7265	7265
R-Square	0.2281	0.2146

Note: The numbers in brackets are t-values; ***, **, and * indicate significance levels at 1 %, 5 %, and 10 %, respectively.

higher education received by corporate executives and the proxy variable for corporate ESG performance is consistently positive and significant at the 5 % level. This indicates that the aforementioned research conclusions are robust.

4.3.2. Excluding the impact of exogenous shocks

Considering the outbreak of the COVID-19 pandemic in China at the end of 2019, which severely impacted economic operations and caused social disarray, this study excludes samples from after 2019 to avoid interference from this sudden public health event on the research conclusions. Based on Model (1), a re-examination was conducted, and the test results, as shown in Table 4, indicate that the regression coefficient between the proxy variable for the level of higher education received by corporate executives and the proxy variable for corporate ESG performance is significantly positive at both the 5 % and 1 % levels, consistent with the aforementioned research conclusions.

4.4. Mechanism test

The educational background of executives has a significant impact on enhancing the level of internal control within companies, thereby promoting corporate ESG (Environmental, Social, Governance) performance. Internal control, as a crucial mechanism for protecting company assets, ensuring the accuracy and legality of financial reports, and facilitating the achievement of corporate objectives, is closely related to ESG performance (Gigante and Manglaviti, 2022). Executives with higher education typically possess better professional qualifications and management abilities, enabling them to design and implement internal control mechanisms more effectively. This can reduce environmental and social risks, improve corporate governance transparency and compliance, and drive ESG performance improvement. Therefore, the educational background of executives indirectly influences corporate ESG performance through enhancing the level of internal controls, potentially aiding companies in achieving better ESG outcomes. Based on this, the paper uses internal control quality data disclosed by Dibo Risk Management Company for Chinese listed companies to measure the internal control level of sample companies. Following the approach of Baron and Kenny (1986), a two-step method is used to test the impact of executives' higher education level on corporate ESG performance, with results shown in Table 5. It is found that the proxy variable for higher education significantly affects the level of internal control at the 1 % level. The results in Column (2) indicate that after adding the proxy variable for internal control quality, the proxy variable for the higher education level of corporate executives remains significantly related to the corporate ESG performance proxy variable at the 10 % level, and the internal control quality proxy variable is significantly positive at the 5 % level. This suggests that internal control quality can partially mediate the effect. Overall, within companies, the role of higher education level in enhancing corporate ESG performance can be indirectly realized through improving the quality of internal controls.

Table 5

Mechanism test.

	Internal control quality	ESG
	(1)	(2)
Degree	0.1365*** (2.6843)	0.0572* (1.8983)
Internal control quality		0.1246** (2.0143)
Constant	0.3358*** (12.8799)	0.3364*** (10.9473)
Controlvariable	Yes	Yes
Year	Yes	Yes
Industry	Yes	Yes
N	17,032	17,032
R-Square	0.1261	0.1754

Note: The numbers in brackets are t-values; ***, **, and * indicate significance levels at 1 %, 5 %, and 10 %, respectively.

Table 6
Ownership heterogeneity.

	ESG	
	non-state-owned enterprises (1)	state-owned enterprises (2)
Degree	0.0014** (1.9821)	0.0615 (1.5083)
Constant	0.5248*** (7.2703)	0.4277*** (7.4115)
Controlvariable	Yes	Yes
Year	Yes	Yes
Industry	Yes	Yes
N	8695	8337
R-Square	0.1905	0.1641

Note: The numbers in brackets are t-values; ***, **, and * indicate significance levels at 1 %, 5 %, and 10 %, respectively.

4.5. Cross-sectional analysis

The aforementioned results of this study revealed a positive correlation between the educational level of corporate executives and corporate ESG (Environmental, Social, Governance) performance. It is worth exploring in depth whether this conclusion varies across companies with different ownership structures. Firstly, the impact of executives' education level on ESG performance may differ between state-owned enterprises (SOEs) and non-state-owned enterprises (non-SOEs). Decision-making in SOEs is influenced by government policies and stakeholders, while non-SOEs may focus more on market-oriented operations and the realization of long-term value. Therefore, the impact of executives' education level on ESG performance might exhibit different effects between these two types of enterprises. Secondly, in enterprises with different ownership structures, the education level of executives may interact with factors such as decision-making transparency and governance structure, thereby affecting ESG performance. SOEs may face pressures from government regulation and public opinion, and the educational level of executives might be related to the enterprise's governance capabilities and responsiveness to external environments. The pathway through which it impacts ESG performance might differ from that in non-SOEs. Thus, the nature of enterprise ownership might modulate the impact of executives' education level on ESG performance. Based on this, the paper groups the study samples by ownership type, with group test results shown in Table 6. It is observed that the positive significant relationship between the proxy variable for executives' higher education level and the proxy variable for corporate ESG performance level only exists in non-SOEs. In SOEs, the regression coefficient of the proxy variable for executives' higher education level is positive but not statistically significant, indicating that the positive impact of executives' higher education level on corporate ESG performance primarily exists within the non-SOEs sample group.

5. Conclusion

The findings of this paper demonstrate that higher education plays a significant role in enhancing corporate ESG (Environmental, Social, Governance) performance. Firstly, it highlights the close relationship between executives' educational background and corporate ESG performance, offering new motivation and direction for the educational training of corporate leadership. Secondly, this discovery underscores the importance of investment in education for the long-term sustainable development of enterprises, providing references for the formulation of human resource development strategies. Finally, the results of this study also offer policy suggestions for governments and the business community to promote the improvement of ESG performance, such as strengthening executive education and training, and establishing ESG performance evaluation mechanisms. These measures can help enhance corporate social responsibility awareness and promote the achievement of sustainable development.

CRedit authorship contribution statement

Kaidi Tu: Writing – review & editing, Visualization, Validation, Methodology, Investigation. **Yuanyuan Guo:** Writing – review & editing, Writing – original draft, Supervision, Software, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

References

- Baron, R.M., Kenny, D.A., 1986. The moderator–mediator variable distinction in social psychological research: conceptual, strategic, and statistical considerations. *J. Pers. Soc. Psychol.* 51 (6), 1173.
- Broadstock, D.C., Chan, K., Cheng, L.T., Wang, X., 2021. The role of ESG performance during times of financial crisis: evidence from COVID-19 in China. *Financ. Res. Lett.* 38, 101716.
- Gao, J., Chu, D., Zheng, J., Ye, T., 2022. Environmental, social and governance performance: can it be a stock price stabilizer? *J. Clean. Prod.* 379, 134705.
- Gigante, G., Manglaviti, D., 2022. The ESG effect on the cost of debt financing: a sharp RD analysis. *Int. Rev. Financ. Anal.* 84, 102382.
- Gillan, S.L., Koch, A., Starks, L.T., 2021. Firms and social responsibility: a review of ESG and CSR research in corporate finance. *J. Corp. Fin.* 66, 101889.
- Li, C., Wu, M., Chen, X., Huang, W., 2022. Environmental, social and governance performance, corporate transparency, and credit rating: some evidence from Chinese A-share listed companies. *Pac. Basin Finance J.* 74, 101806.
- Liu, Y., Zhang, J., Dai, Y., 2023. Analyst following and greenwashing decision[J]. *Financ. Res. Lett.* 58, 104510.
- Nazir, M., Akbar, M., Akbar, A., Poulou, P., Hussain, A., Qureshi, M.A., 2022. The nexus between corporate environment, social, and governance performance and cost of capital: evidence from top global tech leaders. *Environ. Sci. Pollut. Res.* 1–14.
- Tang, H., 2022. ESG performance, investors' heterogeneous beliefs, and cost of equity capital in China. *Front. Environ. Sci.* 10, 992559.
- Wang, W., Yu, Y., Li, X., 2022. ESG performance, auditing quality, and investment efficiency: empirical evidence from China. *Front. Psychol.* 13, 948674.
- Wang, Z., Chu, E., Hao, Y., 2024. Towards sustainable development: how does ESG performance promotes corporate green transformation. *Int. Rev. Financ. Anal.* 91, 102982.
- Wu, Q., Chen, G., Han, J., Wu, L., 2022. Does corporate ESG performance improve export intensity? evidence from Chinese listed firms. *Sustainability.* 14 (20), 12981.
- Zhang, J., Li, Y., Xu, H., Ding, Y., 2023. Can ESG ratings mitigate managerial myopia? Evidence from Chinese listed companies. *Int. Rev. Financ. Anal.* 90, 102878.